**OAK INTERNATIONAL SCHOOL**

EXCELLENCE ENSEMBLE • DUAL CURRICULUM MATHEMATICS

| IB MIDDLE YEARS & DIPLOMA | BENIN NATIONALE APC | BILINGUAL EN / FR |
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## PREMIUM MATHEMATICS ACTIVITY WORKBOOK

# CINQUIÈME (5ÈME) / IB MYP 3 (AGES 12–13)

### A Harmonised Dual-Curriculum Mathematics Learning Companion

*Connecting Conceptual Inquiry, Axiomatic Proof, and Examination Mastery*

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|---|---|
| **Target Cohort:** | Cinquième (5ème) / IB MYP 3 (Ages 12–13) |
| **Academic Year:** | 2026–2027 |
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| **Series Version:** | OIS-WB-5EME-MYP3-V1.0 (Classroom & Exam Certified) |

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**Academic Year:** 2026–2027 | **Series Version:** OIS-WB-5EME-MYP3-V1.0

**Programmes:** IB Middle Years Programme (English) & Beninese National APC Curriculum (Français)

# SECTION A — STUDENT INTRODUCTION / INTRODUCTION POUR L'ÉLÈVE

## Unlocking Geometric Elegance and Algebraic Foundations

*Bienvenue en classe de Cinquième / IB MYP 3 à Oak International School !*

In this pivotal academic year, your mathematical universe expands into negative numbers and formal planar properties:

- In **IB MYP 3**, you investigate general algebraic rules (Criterion B), apply scientific notation to astronomical and atomic scales, and construct real-world linear models for distance, speed, and cost (Criterion D).
- In **Beninese 5ème (APC)**, you explore **Situation d'Apprentissage N°2 (Configurations du plan)** and **N°3 (Nombres et calculs)**. You establish the timeless properties of parallel lines cut by a transversal, prove that the angle sum of any triangle is exactly 180^circ, and uncover the rotational magic of central symmetry (*symétrie centrale*) and parallelograms.

This workbook is your dual-language toolkit to ensure complete mastery of both programmes without cognitive interference.

# SECTION B — BILINGUAL MATHEMATICAL LANGUAGE / GLOSSAIRE BILINGUE

## Angles, Quadrilaterals & Signed Numbers

| English Term | French Equivalent (*Terme Français*) | Mathematical Definition & Symbolic Example | Common Translation Pitfalls |
|---|---|---|---|
| **Directed numbers / Integers** | *Nombres relatifs / Entiers relatifs* | Signed numbers: **Z** = ..., -3, -2, -1, 0, +1, +2, +3, .... | In French, *relatif* includes both positive and negative values. |
| **Alternate interior angles** | *Angles alternes-internes* | Non-adjacent angles on opposite sides of a transversal between parallel lines: hatα = hatβ. | In French: *alternes-internes* (never say "angles alternatifs"). |
| **Corresponding angles** | *Angles correspondants* | Angles in matching relative positions at each intersection of two lines with a transversal. | In French: *angles correspondants*. |
| **Vertically opposite angles** | *Angles opposés par le sommet* | Angles formed by two intersecting lines sharing a common vertex: hat1 = hat2. | In English, often called "vertical angles". |
| **Point reflection / Central symmetry** | *Symétrie centrale* | Transformation through point O where O is the midpoint of [MM']: S_O(M) = M'. | In English, equivalent to a rotation of 180^circ around point O. |
| **Parallelogram** | *Parallélogramme* | A quadrilateral with both pairs of opposite sides parallel and equal. | Diagonals bisect each other (*se coupent en leur milieu*). |
| **Scientific notation** | *Écriture scientifique* | Expression in form a × 10■, where 1 ≤ a < 10 and k ∈ **Z**. | French *notation scientifique* has the identical mathematical constraint on a. |
| **Order of magnitude** | *Ordre de grandeur* | The power of 10 closest to the numerical value: e.g., 3.8 × 10^5 ≈ 10^5 or 10^6. | In French: *ordre de grandeur*. |

# SECTION C — SHARED CONCEPTUAL FOUNDATIONS

## Unit Theme: Parallel Lines, Transversals and Triangle Angle Sums

### 1. Fundamental Theorem of Angles Cut by a Transversal

Let (d_1) and (d_2) be two parallel lines cut by a secant transversal line (Δ):

- Any pair of **alternate interior angles** (*angles alternes-internes*) are equal.
- Any pair of **corresponding angles** (*angles correspondants*) are equal.
- Conversely: If two lines cut by a transversal form equal alternate interior angles, then the lines are strictly parallel.

```
(Δ)
/
-------/------- (d1)
/ alpha
/
----/--------- (d2)
/ beta
/
```

### 2. Universal Proof: The Angle Sum of a Triangle is 180^circ

Consider any triangle ABC:

1. Through vertex A, construct auxiliary line (d) parallel to side (BC).
2. Line (d) forms two angles with sides [AB] and [AC]:
   - Angle hat1 is alternate interior to angle ∠ABC.
   - Angle hat2 is alternate interior to angle ∠ACB.
3. Angles hat1, ∠BAC, and hat2 together form a straight angle on line (d):

hat1 + ∠BAC + hat2 = 180^circ

4. Substituting the equal alternate interior angles:

∠ABC + ∠BAC + ∠ACB = 180^circ

This proof is identical in both the IB inquiry continuum and the Beninese NPE guide!

### 3. Worked Example: Signed Numbers and Precedence

**Problem**: Calculate A = -15 - (-8) + (+12) - (+20).

- **Step 1 (Transform subtractions into additions of opposites)**:

A = -15 + (+8) + (+12) + (-20)

- **Step 2 (Group positive and negative integers)**:

A = [(+8) + (+12)] + [(-15) + (-20)]

A = (+20) + (-35)

- **Step 3 (Calculate final sum)**:

A = -15

## SECTION D — CONTEXTUALISED BRIDGING ACTIVITIES

### Activity: The Tanguiéta Solar Farm Tracker Alignment

#### 1. Problem Situation

An electrical engineering team is installing dual-axis photovoltaic solar arrays in Tanguiéta (Atacora department).

- Rows of solar panels are mounted along parallel support girders (L_1) and (L_2).
- A sun-tracking mechanical linkage rod (T) crosses girders (L_1) and (L_2) at pivot points P and Q.
- A sensor at pivot P records an interior angle between girder (L_1) and linkage (T) of 64^circ.
- A robotic actuator at pivot Q must be calibrated to position the lower panel at angle $\theta$ such that the panels remain parallel to prevent shadowing.

#### 2. The IB Perspective (Inquiry into Geometric Models)

- Identify (L_1) and (L_2) as parallel lines.
- Identify linkage (T) as the transversal.
- By geometric angle theorems, the alternate interior angle at Q must equal 64^circ.
- If the actuator measures the co-interior (interior on the same side) angle $\phi$:

$\phi$ = 180^circ - 64^circ = 116^circ

#### 3. The Beninese APC Perspective (Structured Proof / *Démonstration*)

- *Données* : Les poutres de support (L_1) et (L_2) sont parallèles : (L_1) ■ (L_2). La tige sécante (T) coupe (L_1) en P et (L_2) en Q. L'angle widehatL_1 P Q = 64^circ.
- *Propriété* : Si deux droites parallèles sont coupées par une sécante, alors les angles alternes-internes qu'elles forment ont la même mesure.
- *Calcul et Conclusion* : L'angle alterne-interne widehatP Q L_2 et l'angle widehatL_1 P Q sont de même mesure. Donc $\theta$ = 64^circ.
- *Vérification supplémentaire* : Les angles intérieurs d'un même côté sont supplémentaires : 64^circ + 116^circ = 180^circ.

#### 4. Comparison of Methods

- Both programmes arrive at the exact same angular values (64^circ and 116^circ).
- The Beninese APC demands an explicit reference to the statutory property name (*angles alternes-internes*), while the IB emphasizes functional sensor orientation.

#### 5. Student Reflection Question

*Why is it physically impossible for solar panels mounted on parallel girders to have unequal alternate interior angles without bending the linkage rod?*

*Student Working Space / Espace réservé à la rédaction de l'élève :*

#### 6. Application in a New Context

In an urban development in Abomey-Calavi, two parallel avenues are crossed diagonally by a railway line at an angle of 72^circ. Determine all eight angles formed at the two intersections.

*Student Working Space / Espace réservé à la rédaction de l'élève :*

# SECTION E — PROGRESSIVE PRACTICE / EXERCICES PROGRESSIFS

## Level 1: Foundational Understanding (Signed Numbers & Index Laws)

**1.** Calculate:

a) (-14) + (+9) =

b) (-8) - (-15) =

c) (-6) × (-7) =

d) (+48) ÷ (-8) =

**2.** Write in scientific notation (a × 10■):

a) 450 000 =

b) 0.000 78 =

c) The speed of light: 300 000 000 m/s =

d) 125 × 10■ =

## Level 2: Standard Application (Angles & Fractions with Unlike Denominators)

**3.** Calculate and give the irreducible fraction:

a) (3 / 4) + (5 / 6) =

b) (7 / 10) - (4 / 15) =

c) (5 / 8) × (16 / 25) =

**4.** In triangle KLM, angle hatK = 58^circ and angle hatL = 74^circ.

a) Calculate the measure of angle hatM.

b) Is triangle KLM acute, right-angled, or obtuse? Justify.

## Level 3: Multi-Step Problems (Central Symmetry & Parallelograms)

**5.** Let ABCD be a parallelogram with centre O.

a) What is the image of vertex A under the central symmetry S_O?

b) What is the image of segment [AB] under S_O?

c) Prove that line (AB) is parallel to line (CD) using the properties of central symmetry.

**6. Proportionality and Journey Rates**:

A refrigerated truck carrying fish from Cotonou to Malanville travels 720 km.

- The truck maintains a constant average speed of 60 km/h.
- Fuel consumption is directly proportional to distance: 18 litres per 100 km.

a) Calculate the total travel time in hours.

b) Calculate the total volume of diesel consumed.

c) If diesel costs 650 CFA F per litre, calculate the total fuel expenditure.

## Level 4: Reasoning & Geometric Proof

**7.** In the figure below, lines (AB) and (CD) are parallel. Triangle CDE is isosceles at C.

Angle ∠BAC = 45^circ and angle ∠ACD = 45^circ.

Line (AC) is a transversal.

Angle ∠CDE = 70^circ.

a) Prove that triangle ACD is right-angled if ∠ADC = 45^circ.

b) Calculate the measures of all three interior angles of triangle CDE.

c) Write a structured proof for each deduced angle.

## Level 5: Extension & 5ème National Examination Challenge

**8. Situation d'Évaluation Type 5ème (SA N°2 & N°4)**:

Pour sécuriser un domaine agricole à Savè, un géomètre implante quatre bornes M, N, P, Q.

Il vérifie sur le terrain que le point I est le milieu commun des segments [MP] et [NQ].

De plus, il mesure MN = 145 m et ∠MNP = 65^circ.

1. Fais une figure géométrique soignée représentant cette situation (échelle 1 cm pour 20 m).
2. Démontre que le quadrilatère MNPQ est un parallélogramme.
3. Déduis-en la longueur exacte du côté [PQ] sans nouvelle mesure.
4. Démontre que les droites (MN) et (PQ) sont strictement parallèles.
5. Détermine la mesure de l'angle ∠NPQ.

## SECTION F — INTEGRATED APPLICATION TASKS

### Task F-1: The Songhaï Bio-Gas Digester Network (MYP Criterion D & Benin SA 5)

**Context**: The Songhaï Agricultural Innovation Centre in Porto-Novo generates renewable bio-gas from agricultural biomass to power a collective bakery and dairy processing unit.

- The underground main gas pipe runs along straight line (G_1).
- A branch line (G_2) is laid parallel to (G_1).
- A cross-connecting pipe (C) intersects (G_1) at valve V_1 and (G_2) at valve V_2.
- At valve V_1, the connecting pipe forms an acute angle of 52^circ with line (G_1).
- The pipe length V_1 V_2 = 85 metres.

**Questions**:

1. Draw a schematic diagram of the pipe network labelling (G_1), (G_2), and transversal (C).
2. State the measure of the alternate interior angle at valve V_2 and justify using the appropriate theorem.
3. Determine the obtuse angle formed by the cross-connector (C) and line (G_2).
4. Gas flows through the cross-connector at an average speed of 2.5 m/s. Calculate how many seconds it takes for gas to travel from V_1 to V_2.
5. If the daily gas production is 450 m³ and the bakery consumes (5 / 9) of this production, calculate the volume of gas remaining for the dairy unit.

# SECTION G — PROGRAMME-SPECIFIC REINFORCEMENT

## Part G-1: IB MYP 3 Criterion B Pattern Investigation

**Investigation: Consecutive Angle Bisector Grids**

Consider a sequence of regular polygons:

- Equilateral triangle (3 sides): Interior angle sum = 180^circ.
- Quadrilateral (4 sides): Interior angle sum = 360^circ.
- Pentagon (5 sides): Interior angle sum = 540^circ.

**1.** Record values in a table for polygons with n = 3, 4, 5, 6, 7, 8 sides.

**2.** Formulate a general rule for the sum of interior angles S in terms of number of sides n.

**3.** Test and justify your general rule using triangulation from a single vertex.

**4.** Calculate the interior angle sum of a polygon with 24 sides (icositetragon).

## Part G-2: Beninese NPE Formal Proof Exercise

**Exercice de Démonstration Formelle (Classe de 5ème)**

Soit un triangle ABC. Soit I le milieu de [AB] et J le milieu de [AC].

Soit K le symétrique de I par rapport au point J.

**1.** Démontre que le quadrilatère AICK est un parallélogramme.

**2.** Déduis-en que les segments [CK] et [AI] sont parallèles et de même longueur.

**3.** Démontre ensuite que le quadrilatère BCKI est un parallélogramme.

*Rédige rigoureusement chaque étape avec les propriétés de la symétrie centrale.*

# SECTION H — ASSESSMENT, REFLECTION & WORKED SOLUTIONS

## Diagnostic Pre-Flight Check

**[ ]** I can add and subtract negative numbers with confidence.
**[ ]** I can find a common denominator for two fractions (e.g., 4 and 6 → 12).
**[ ]** I know that alternate interior angles between parallel lines are equal.
**[ ]** I know that the diagonals of a parallelogram bisect each other.

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## COMPLETE WORKED SOLUTIONS & MARKING SCHEME

### Solutions for Section E (Progressive Practice)

**1. Signed Operations** (1 mark each):

a) -14 + 9 = -5

b) -8 + 15 = +7

c) (-6) × (-7) = +42

d) 48 ÷ (-8) = -6

**2. Scientific Notation** (1 mark each):

a) 4.5 × 10^5

b) 7.8 × 10^-4

c) 3.0 × 10^8 m/s

d) 1.25 × 10^6

**3. Fractions Unlike Denominators** (1 mark each):

a) (3 × 3 / 12) + (5 × 2 / 12) = (9+10 / 12) = (19 / 12).

b) (7 × 3 / 30) - (4 × 2 / 30) = (21-8 / 30) = (13 / 30).

c) (5 × 16 / 8 × 25) = (1 × 2 / 1 × 5) = (2 / 5).

**4. Triangle Angles** (2 marks):

a) hatM = 180^circ - (58^circ + 74^circ) = 180^circ - 132^circ = 48^circ.

b) All three angles are strictly less than 90^circ (58^circ, 74^circ, 48^circ < 90^circ); therefore, triangle KLM is an acute-angled triangle (*triangle acutangle*).

**5. Parallelogram Central Symmetry** (3 marks):

a) Under S_O, the image of A is C (since O is the midpoint of [AC]).

b) Under S_O, the image of B is D. Thus, the image of segment [AB] is segment [CD].

c) Central symmetry transforms any straight line into a line parallel to itself. Therefore, line (AB) is strictly parallel to line (CD).

**6. Refrigerated Truck** (3 marks):

a) Travel time: t = (d / v) = (720 / 60) = 12 hours.

b) Diesel volume: (720 × 18 / 100) = 7.2 × 18 = 129.6 litres.

c) Expenditure: 129.6 × 650 = 84 240 CFA F.

**7. Angle Reasoning** (4 marks):

a) In triangle ACD, angles are 45^circ, 45^circ. Sum = 90^circ. Remaining angle ∠ADC = 180^circ - 90^circ = 90^circ. Right-angled at D.

b) In triangle CDE isosceles at C, base angles ∠CDE = ∠CED = 70^circ.

Vertex angle ∠DCE = 180^circ - (70^circ + 70^circ) = 40^circ.

**8. Savè Farm 5ème Task** (5 marks):

1. Scale diagram (145 m → 7.25 cm). [1 mark]
2. Segments [MP] and [NQ] have the same midpoint I. A quadrilateral whose diagonals bisect each other is a parallelogram. Thus, MNPQ is a parallelogram. [2 marks]
3. In a parallelogram, opposite sides are equal in length. Hence PQ = MN = 145 m. [1 mark]
4. In a parallelogram, opposite sides are parallel. Hence (MN) ■ (PQ). [1 mark]

### Solutions for Section F (Task F-1: Songhaï Bio-Gas)

1. Labeled schematic diagram. [1 mark]
2. Alternate interior angle at V_2 = 52^circ, because (G_1) ■ (G_2) cut by transversal (C). [2 marks]
3. Obtuse supplementary angle: 180^circ - 52^circ = 128^circ. [1 mark]

4. Transit time: t = (d / v) = (85 / 2.5) = 34 seconds. [2 marks]
5. Gas remaining: (5 / 9) × 450 = 250 m³ used by bakery. Remaining = 450 - 250 = 200 m³. [2 marks]

### Solutions for Section G-2 (Benin Formal Proof 5ème)

1. *Démonstration AICK* : J est le milieu de [AC] (par hypothèse) et J est le milieu de [IK] (car K est le symétrique de I par rapport à J). Le quadrilatère AICK a ses diagonales [AC] et [IK] qui se coupent en leur milieu J ; c'est donc un parallélogramme. [2 pts]
2. *Conséquence* : Puisque AICK est un parallélogramme, ses côtés opposés [CK] et [AI] sont parallèles et de même longueur : (CK) ■ (AI) et CK = AI. [1 pt]
3. *Démonstration BCKI* : On sait que I est le milieu de [AB], donc AI = IB, et les points A, I, B sont alignés. Comme CK = AI, on a aussi CK = IB. De plus, comme (CK) ■ (AI) et que B ∈ (AI), on a (CK) ■ (IB). Le quadrilatère BCKI possède deux côtés opposés [CK] et [IB] qui sont parallèles et de même longueur ; c'est donc un parallélogramme. [2 pts]

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*End of Workbook 3 (Cinquième / MYP 3).*